

CF Game User Testing Protocol (v3)

Quick Pitch (to read to participant)

This tool helps you evaluate counterfactual explanations generated by AI models. You will first make your own judgment about which explanation is best, based only on what you see. After that, the system reveals evaluation metrics such as validity and plausibility. The goal is to explore how your intuition compares to these metrics, and whether the interface helps you understand any differences between them.

1. Objective

Evaluate how human judgments of counterfactual explanations align with objective metrics, and how the structured dashboard helps reveal or reduce discrepancies.

2. Method

Think-aloud usability test followed by a short probing interview.

3. Setup

Quiet environment, screen recording if possible, one facilitator and one note-taker, obtain consent.

4. Instructions to Participant

You will use a prototype showing counterfactual explanations. Please think aloud continuously. There are no right or wrong answers. We are testing the system, not you.

5. Task

Complete one full run of the CF Game from initial judgment to final reflection.

6. Observation Guidelines

Record what the participant says and does, confusion moments, key decisions, and any assistance provided.

7. Post-Test Interview Questions

Q1. Decision Process

Can you explain step by step why you selected this counterfactual as the best one?

Q2. Alignment with Metrics

When you saw the metrics, did they match your expectations? What surprised you the most?

Q3. Interface Understanding

At what point in the interface did you understand the difference between your judgment and the metrics, and what helped or did not help?

Q4. Structured Dashboard Improvement (UPDATED)

If you could change one thing in the structured dashboard (e.g. layout, order of steps, or how metrics are shown) to better understand the difference between your judgment and the metrics, what would it be?

Follow-up: Why would that help you?

8. General Follow-up Prompt

Can you give a concrete example?

9. Data to Capture per Participant

Initial choice, expected best CF, metric-best CF, whether choice changed, confusion moments, key quotes.

10. Outcome

Use findings to quantify alignment and derive concrete interface improvements.